

ECE406 - HOMEWORK 2

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$$1) \begin{bmatrix} a_0 \\ b_0 \end{bmatrix} = \begin{bmatrix} \sqrt{h} & -i\sqrt{1-h} \\ -i\sqrt{1-h} & \sqrt{h} \end{bmatrix} \begin{bmatrix} a_0 \\ b_0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} \sqrt{h} & i\sqrt{1-h} \\ i\sqrt{1-h} & \sqrt{h} \end{bmatrix} \begin{bmatrix} a_0 \\ b_0 \end{bmatrix}$$

$$|\alpha\rangle_{a_1} |\beta\rangle_{b_1} = D_{a_1}(\alpha) D_{b_1}(\beta) |0\rangle_{a_1} |0\rangle_{b_1} = e^{\alpha a_1^\dagger - \alpha^* a_1} e^{\beta b_1^\dagger - \beta^* b_1} |0\rangle_{a_1} |0\rangle_{b_1}$$

$$= e^{\alpha(\sqrt{h} a_0^\dagger - i\sqrt{1-h} b_0^\dagger) - \alpha^*(\sqrt{h} a_0 + i\sqrt{1-h} b_0)} e^{\beta(-i\sqrt{1-h} a_0^\dagger + \sqrt{h} b_0^\dagger) - \beta^*(i\sqrt{1-h} a_0 + \sqrt{h} b_0)} |0\rangle_{a_1} |0\rangle_{b_1}$$

* Since a_0, b_0 terms commute we can: $e^{A(a_0) + B(b_0)} = e^{A(a_0)} e^{B(b_0)}$

$$|\alpha\rangle_{a_1} |\beta\rangle_{b_1} = e^{\sqrt{h}(\alpha a_0^\dagger - \alpha^* a_0)} e^{-i\sqrt{1-h}(\alpha b_0^\dagger - \alpha^* b_0)} e^{-i\sqrt{1-h}(\beta a_0^\dagger + \beta^* a_0)} e^{\sqrt{h}(\beta b_0^\dagger - \beta^* b_0)} |0\rangle_{a_1} |0\rangle_{b_1}$$

$$= e^{\sqrt{h}(\alpha a_0^\dagger - \alpha^* a_0)} e^{-i\sqrt{1-h}(\beta a_0^\dagger + \beta^* a_0)} e^{-i\sqrt{1-h}(\alpha b_0^\dagger + \alpha^* b_0)} e^{\sqrt{h}(\beta b_0^\dagger - \beta^* b_0)} |0\rangle_{a_1} |0\rangle_{b_1}$$

→ We've switched the order of 2nd and 3rd exponentials above, since they commute

$$|\alpha\rangle_{a_1} |\beta\rangle_{b_1} = e^A e^B e^C e^D \quad \text{where} \quad \begin{aligned} A &= \sqrt{h}(\alpha a_0^\dagger - \alpha^* a_0) \\ B &= -i\sqrt{1-h}(\beta a_0^\dagger + \beta^* a_0) \\ C &= -i\sqrt{1-h}(\alpha b_0^\dagger + \alpha^* b_0) \\ D &= \sqrt{h}(\beta b_0^\dagger - \beta^* b_0) \end{aligned}$$

Recall that, from BCH formula:

$$e^X e^Y = e^Z, \quad Z = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \dots$$

$$[A, B] = -i\sqrt{h(1-h)} [\alpha\beta^*[a_0^\dagger, a_0] - \alpha^*\beta[a_0, a_0^\dagger]]$$

$$= +i\sqrt{h(1-h)} (\alpha\beta^* + \alpha^*\beta)$$

* Notice that $[a_0^\dagger, a_0^\dagger] = [a_0, a_0] = 0$ trivially.

$$\text{Similarly } [C, D] = -i\sqrt{h(1-h)} \{-\alpha\beta^*[b_0^\dagger, b_0] + \alpha^*\beta[b_0, b_0^\dagger]\} \\ = -i\sqrt{h(1-h)} (\alpha\beta^* + \alpha^*\beta) = -[A, B]$$

→ $[A, [A, B]] = [C, [C, D]] = 0$ since first commutators are just numbers,

$$|\alpha\rangle_{a_i} |\beta\rangle_{b_i} = e^{A+B} e^{\frac{1}{2}[A, B]} e^{C+D} e^{\frac{1}{2}[C, D]} |0\rangle_{a_i} |0\rangle_{b_i} = e^{A+B} e^{C+D} |0\rangle_{a_i} |0\rangle_{b_i}$$

Noticing that $|0\rangle_{a_i} |0\rangle_{b_i} \rightarrow |0\rangle_{a_0} |0\rangle_{b_0}$, output state becomes

$$e^{\underbrace{(\alpha\sqrt{h} - i\beta\sqrt{1-h})}_{\gamma} a_0^\dagger - \underbrace{(\alpha^*\sqrt{h} + i\beta^*\sqrt{1-h})}_{\gamma^*} a_0} \\ \times e^{\underbrace{(\beta\sqrt{h} - i\alpha\sqrt{1-h})}_{\xi} b_0^\dagger - \underbrace{(\beta^*\sqrt{h} + i\alpha^*\sqrt{1-h})}_{\xi^*} b_0} |0\rangle_{a_0} |0\rangle_{b_0}$$

$$= D_{a_0}(\gamma) D_{b_0}(\xi) |0\rangle_{a_0} |0\rangle_{b_0} = |\gamma\rangle_{a_0} |\xi\rangle_{b_0}, \quad \begin{array}{l} \gamma = \alpha\sqrt{h} - i\beta\sqrt{1-h} \\ \xi = \beta\sqrt{h} - i\alpha\sqrt{1-h} \end{array}$$

→ Intuitively BS is

$$\begin{bmatrix} \gamma \\ \xi \end{bmatrix} = \begin{bmatrix} \cos\theta & -i\sin\theta \\ -i\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow \text{Rotation around } \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ in } SU(2) \\ \text{where } \cos\theta = \sqrt{h}, \sin\theta = \sqrt{1-h} \\ \hookrightarrow \text{Unitary operation!}$$

→ Output state is a product state and hence is not entangled like the single photon case. This means coherent states interfere just like the classical light.

* Acknowledgement: Wikipedia page of Beam Splitter was used to make the intuitive comment above

$$1.b) \text{ input state} = a_i^\dagger D_{b_i}(\beta) |0\rangle_{a_i} |0\rangle_{b_i}$$

$$= (\sqrt{\hbar} a_0^\dagger - i\sqrt{1-\hbar} b_0^\dagger) e^{\beta(-i\sqrt{1-\hbar} a_0^\dagger + \sqrt{\hbar} b_0^\dagger) - \beta^*(i\sqrt{1-\hbar} a_0 + \sqrt{\hbar} b_0)} |0\rangle_{a_i} |0\rangle_{b_i}$$

$$= (\sqrt{\hbar} a_0^\dagger - i\sqrt{1-\hbar} b_0^\dagger) e^{-i\sqrt{1-\hbar}(\beta a_0^\dagger + \beta^* a_0)} e^{\sqrt{\hbar}(\beta b_0^\dagger - \beta^* b_0)} |0\rangle_{a_i} |0\rangle_{b_i}$$

→ Since a_0, b_0 commute

$$= (\sqrt{\hbar} a_0^\dagger - i\sqrt{1-\hbar} b_0^\dagger) D_{a_0}(-i\sqrt{1-\hbar}\beta) D_{b_0}(\sqrt{\hbar}\beta)$$

→ Recall that $D^\dagger(\alpha) a^\dagger D(\alpha) = a^\dagger + \alpha^*$. Since D is unitary

$$D D^\dagger a^\dagger D = a^\dagger D(\alpha) = D(\alpha)(a^\dagger + \alpha^*)$$

→ Also since $a_0^\dagger, D_{b_0}$ commute and $b_0^\dagger, D_{a_0}$ commute, state is

$$D_{a_0}(-i\sqrt{1-\hbar}\beta) D_{b_0}(\sqrt{\hbar}\beta) \left[\sqrt{\hbar} (a_0^\dagger + \underbrace{i\sqrt{1-\hbar}\beta}_{\alpha^*}) - i\sqrt{1-\hbar} (b_0^\dagger + \underbrace{\sqrt{\hbar}\beta}_{\beta^*}) \right] |0\rangle_{a_i} |0\rangle_{b_i}$$

$$= D_{a_0}(-i\sqrt{1-\hbar}\beta) D_{b_0}(\sqrt{\hbar}\beta) (\sqrt{\hbar} a_0^\dagger - i\sqrt{1-\hbar} b_0^\dagger) |0\rangle_{a_0} |0\rangle_{b_0}$$

$$= \sqrt{\hbar} D_{a_0}(-i\sqrt{1-\hbar}\beta) D_{b_0}(\sqrt{\hbar}\beta) |1\rangle_{a_0} |0\rangle_{b_0} - i\sqrt{1-\hbar} D_{a_0}(-i\sqrt{1-\hbar}\beta) D_{b_0}(\sqrt{\hbar}\beta) |0\rangle_{a_0} |1\rangle_{b_0}$$

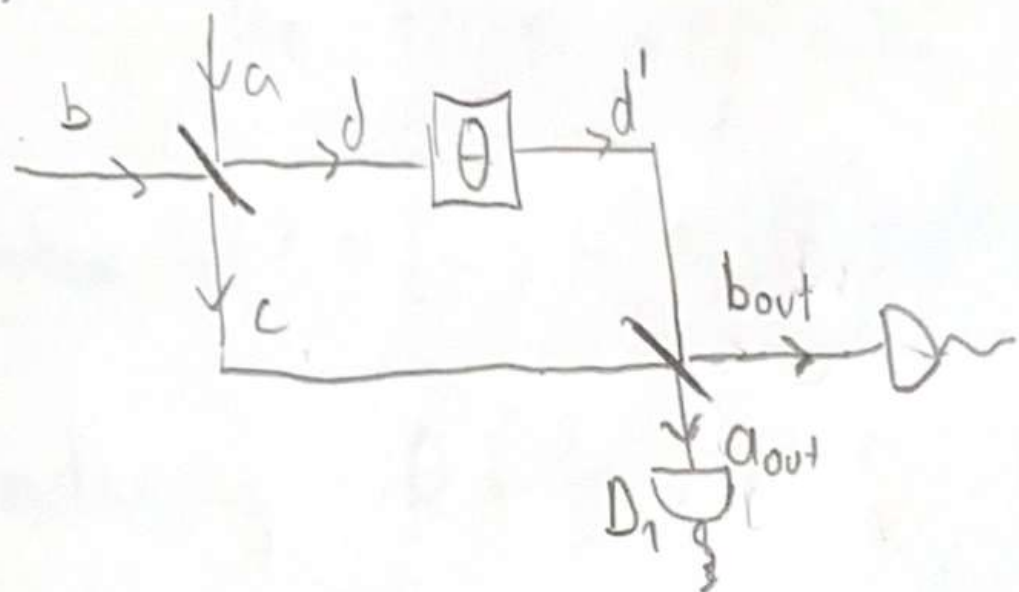
$$= \sqrt{\hbar} \underbrace{|-i\sqrt{1-\hbar}\beta, 1\rangle_{a_0}}_{\text{Displaced Fock State}} \otimes \underbrace{|\sqrt{\hbar}\beta\rangle_{b_0}}_{\text{Displaced Fock State}} - i\sqrt{1-\hbar} \underbrace{|-i\sqrt{1-\hbar}\beta\rangle_{a_0}}_{\text{Displaced Fock State}} \otimes \underbrace{|\sqrt{\hbar}\beta, 1\rangle_{b_0}}_{\text{Displaced Fock State}}$$

Displaced Fock State

Displaced Fock State

* Result is a superposition of product of coherent and displaced Fock states

2)



* Instead of evolving the state let us evolve the operators
Each letter below corresponds to the annihilation op. at different stages of MZI

$$\begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} \sqrt{T} & i\sqrt{R} \\ i\sqrt{R} & \sqrt{T} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}, \quad \begin{bmatrix} c \\ d' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}, \quad \begin{bmatrix} a_{out} \\ b_{out} \end{bmatrix} = \begin{bmatrix} i\sqrt{R} & \sqrt{T} \\ \sqrt{T} & i\sqrt{R} \end{bmatrix} \begin{bmatrix} c \\ d' \end{bmatrix}$$

$$\begin{bmatrix} a_{out} \\ b_{out} \end{bmatrix} = \begin{bmatrix} \sqrt{T} & i\sqrt{R} \\ i\sqrt{R} & \sqrt{T} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} \begin{bmatrix} i\sqrt{R} & \sqrt{T} \\ \sqrt{T} & i\sqrt{R} \end{bmatrix} = \begin{bmatrix} i\sqrt{RT}(1+e^{i\theta}) & (Te^{i\theta}-R) \\ T-Re^{i\theta} & i\sqrt{RT}(1+e^{i\theta}) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$P(D_1=1 \text{ photon}, D_2=2 \text{ photon}) = \langle \Psi_{in} | \hat{\Pi}_{1,2} | \Psi_{in} \rangle$$

→ The projection operator is: $\hat{\Pi}_{1,2} = |1,2\rangle_{out} \langle 1,2|_{out}$

$$\hat{\Pi}_{1,2} = \left(\frac{1}{\sqrt{1,2!}} \right)^2 a_{out}^\dagger |b_{out}^\dagger|^2 |0,0\rangle_{out} \langle 0,0|_{out} a_{out} b_{out}^2$$

$$P(1,2) = \left| \langle 1,2 | \frac{1}{\sqrt{2}} b_{in} a_{in}^2 \right| \left(\frac{1}{2} a_{out}^\dagger |b_{out}^\dagger|^2 |0,0\rangle_{out} \langle 0,0| b_{out}^2 a_{out} \right) \left| \frac{1}{\sqrt{2}} a_{in}^\dagger b_{in}^\dagger |0,0\rangle \right|^2$$

$$= \left| \frac{1}{\sqrt{2}} \langle 0,0|_{out} b_{out} a_{out}^2 |1,2\rangle_{in} \right|^2$$

$$a_{out} b_{out}^2 = \left[i\sqrt{RT}(1+e^{i\theta}) a_{in} + (Te^{i\theta}-R) b_{in} \right] \left[(T-Re^{i\theta}) a_{in} + i\sqrt{RT}(1+e^{i\theta}) b_{in} \right]^2$$

* Note in the product $a_{out} b_{out}^2 |1\rangle_{a_{in}} |2\rangle_{b_{in}}$; terms containing $a_{in}^3, b_{in}^3, a_{in}^2 b_{in}$ will not survive.

Only the terms with $a_{in} b_{in}^2$ will survive

$$a_{out} b_{out}^2 |1,2\rangle = \left[2 (T e^{i\theta} - R) (T - R e^{i\theta}) i \sqrt{RT} (1 + e^{i\theta})^{3/2} (RT)^{3/2} (1 + e^{i\theta})^3 \right] a_{in} b_{in}^2 |1,2\rangle$$

Replacing $R = \frac{2}{3}$, $T = \frac{1}{3}$, $\theta = 0$

$$a_{out} b_{out}^2 |1,2\rangle = \left[i 2 \left(-\frac{1}{3}\right) \left(-\frac{1}{3}\right) \frac{2\sqrt{2}}{3} - i \frac{2\sqrt{2}}{27} \times 8 \right] a_{in} b_{in}^2 |1,2\rangle = -i \frac{4\sqrt{2}}{9} a_{in} b_{in}^2 |1,2\rangle$$

$$a_{in} b_{in}^2 |1,2\rangle = \sqrt{2} |0,0\rangle_{in}$$

$$P(1,2) = \left| -i \frac{4\sqrt{2}}{9} \underbrace{\langle 0,0 | 0,0 \rangle_{in}}_1 \right|^2 = \boxed{\frac{32}{81}}$$

$$3) |\Psi\rangle = N a^\dagger |\alpha\rangle$$

$$g^{(2)}(x_1, x_2; x_2, x_1) = \frac{G^{(2)}(x_1, x_2; x_2, x_1)}{G^{(1)}(x_1, x_1) G^{(1)}(x_2, x_2)}$$

To find $g^2(0)$, we set $x_1 = (r, t) = x_2 = x$

$$g^{(2)}(0) = \frac{\langle E^{(-)}(x) E^{(-)}(x) E^{(+)}(x) E^{(+)}(x) \rangle}{\langle E^{(-)}(x) E^{(+)}(x) \rangle \langle E^{(-)}(x) E^{(+)}(x) \rangle}$$

→ Assuming single mode, we can replace $E^{(-)}(x) = a^\dagger$, $E^{(+)}(x) = a$

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger a a \rangle}{\langle a^\dagger a \rangle^2}$$

$$\langle \Psi | \Psi \rangle = |N|^2 \langle \alpha | a a^\dagger | \alpha \rangle = |N|^2 \langle \alpha | a^\dagger a + 1 | \alpha \rangle$$

* First normalize $|\Psi\rangle \Rightarrow$

$$|N|^2 = \frac{1}{|\alpha|^2 + 1}$$

$$\Rightarrow \langle \Psi | a^\dagger a^\dagger a a | \Psi \rangle = |N|^2 \langle \alpha | a a^\dagger a a^\dagger | \alpha \rangle$$

$$= |N|^2 \langle \alpha | (a^\dagger a + 1) a^\dagger a (a^\dagger a + 1) | \alpha \rangle = \langle \alpha | a^\dagger a a^\dagger a a^\dagger a + 2(a^\dagger a)^2 + a^\dagger a | \alpha \rangle$$

$$= |N|^2 [\langle \alpha | a^\dagger (a^\dagger a + 1) (a^\dagger a + 1) a | \alpha \rangle + 2 \langle \alpha | a^\dagger a a^\dagger a | \alpha \rangle + |\alpha|^2]$$

$$= |N|^2 [\langle \alpha | a^\dagger a^\dagger a a^\dagger a a | \alpha \rangle + 2 \langle \alpha | a^\dagger a^2 a | \alpha \rangle + |\alpha|^2 + 2|\alpha|^4 + 2|\alpha|^2 + |\alpha|^2]$$

$$= |N|^2 [|\alpha|^6 + |\alpha|^4 + 2|\alpha|^4 + 2|\alpha|^4 + 4|\alpha|^2] = \frac{|\alpha|^2 (|\alpha|^4 + 5|\alpha|^2 + 4)}{|\alpha|^2 + 1}$$

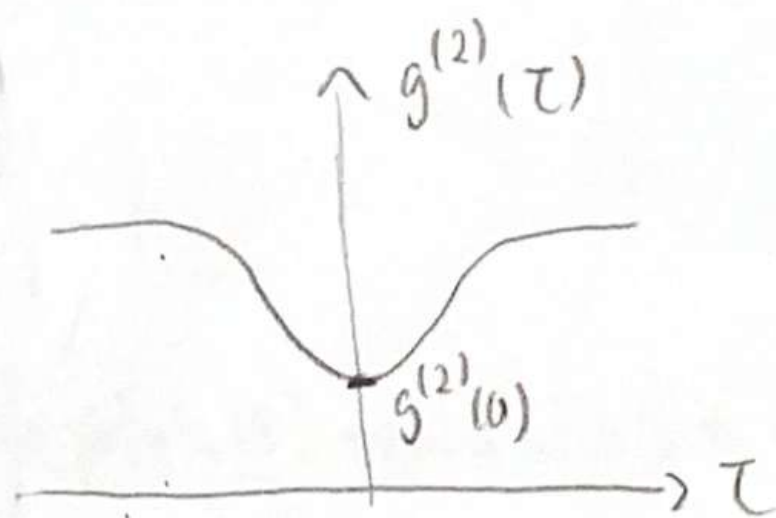
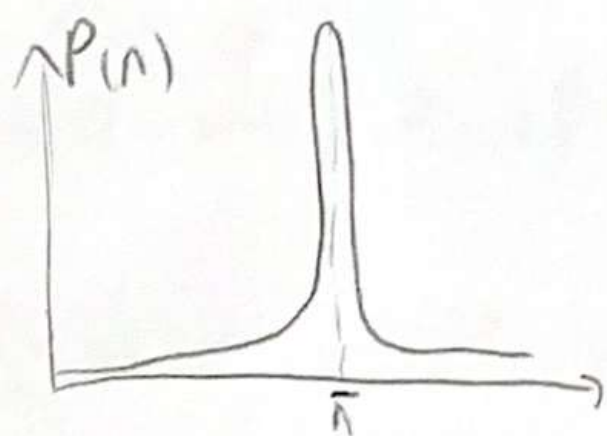
$$\langle a^\dagger a^\dagger a a \rangle = |\alpha|^2 (|\alpha|^2 + 4) = |\alpha|^4 + 4|\alpha|^2$$

$$\begin{aligned}
 \langle a^\dagger a \rangle &= |N|^2 \langle \alpha | a a^\dagger a a^\dagger | \alpha \rangle = |N|^2 \langle \alpha | (a^\dagger a + 1) (a^\dagger a + 1) | \alpha \rangle \\
 &= |N|^2 [\langle \alpha | a^\dagger a a^\dagger a | \alpha \rangle + 2 \langle \alpha | a^\dagger a | \alpha \rangle + 1] \\
 &= |N|^2 [|\alpha|^4 + 3|\alpha|^2 + 1] = \frac{|\alpha|^4 + 3|\alpha|^2 + 1}{|\alpha|^2 + 1}
 \end{aligned}$$

$$g^{(2)}(0) = \frac{|\alpha|^4 + 4|\alpha|^2}{\left(\frac{|\alpha|^4 + 3|\alpha|^2 + 1}{|\alpha|^2 + 1} \right)^2}$$

$$\begin{aligned}
 \text{Letting } |\alpha|^2 = n \Rightarrow g^{(2)}(0) &= \frac{(n^2 + 4n)(n^2 + 2n + 1)}{n^4 + 9n^2 + 1 + 6n^3 + 2n^2 + 6n} \\
 &= 1 - \frac{2n^2 + 2n + 1}{n^4 + 6n^3 + 11n^2 + 6n + 1}
 \end{aligned}$$

* $\underline{g^{(2)}(0) < 1}$, meaning state exhibits sub-Poissonian (anti-bunched) statistics.



* In the limiting case $n = |\alpha|^2 \rightarrow 0$

$g^{(2)}(0) = 0$ and $|\Psi\rangle = N a^\dagger |0\rangle = |1\rangle \Rightarrow$ single-photon !

* In the other limit $n \rightarrow \infty$

$g^{(2)}(0) = 1$ and $|\Psi\rangle \approx |\alpha\rangle \Rightarrow$ almost coherent state !

$$4) \quad |\Psi\rangle_{\pm} = \frac{1}{\sqrt{2}} \left(|\beta\rangle_{\pm} |-\beta\rangle \right)$$

Similar to last question: $g^{(2)}(0) = \frac{\langle a^{\dagger} a^{\dagger} a a \rangle}{\langle a^{\dagger} a \rangle^2}$

$$\begin{aligned} \langle \Psi_{\pm} | a^{\dagger} a^{\dagger} a a | \Psi_{\pm} \rangle &= \frac{1}{2} (\langle \beta | \pm \langle -\beta |) | a^{\dagger} a^{\dagger} a a | (|\beta\rangle \pm |-\beta\rangle) \\ &= \frac{1}{2} [\langle \beta | a^{\dagger} a^{\dagger} a a | \beta \rangle + \langle -\beta | a^{\dagger} a^{\dagger} a a | -\beta \rangle \pm \langle \beta | a^{\dagger} a^{\dagger} a a | -\beta \rangle \pm \langle -\beta | a^{\dagger} a^{\dagger} a a | \beta \rangle] \\ &= \frac{|\beta|^4}{2} [\langle \beta | \beta \rangle + \langle -\beta | -\beta \rangle \pm \langle \beta | -\beta \rangle \pm \langle -\beta | \beta \rangle] \end{aligned}$$

* Inner prod. of any 2 coherent states can be calculated by:

$$\begin{aligned} \langle \alpha | \gamma \rangle &= e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} \langle n | e^{-|\gamma|^2/2} \sum_m \frac{\gamma^m}{\sqrt{m!}} | m \rangle = e^{-\frac{|\alpha|^2 + |\gamma|^2}{2}} \sum_n \frac{(\alpha^* \gamma)^n}{n!} \\ &= \exp\left(-\frac{1}{2} |\alpha|^2 - \frac{1}{2} |\gamma|^2 + \alpha^* \gamma\right) \end{aligned}$$

Then $\langle \beta | -\beta \rangle = e^{-2|\beta|^2}$

$$\langle a^{\dagger} a^{\dagger} a a \rangle = |\beta|^4 \left(1 \pm e^{-2|\beta|^2} \right)$$

Similarly $\langle \Psi_{\pm} | a^{\dagger} a | \Psi_{\pm} \rangle = \frac{1}{2} [\langle \beta | a^{\dagger} a | \beta \rangle + \langle -\beta | a^{\dagger} a | -\beta \rangle \pm \langle \beta | a^{\dagger} a | -\beta \rangle \pm \langle -\beta | a^{\dagger} a | \beta \rangle]$

$$= \frac{1}{2} [\langle \beta | \beta \rangle + \langle -\beta | -\beta \rangle \mp \langle \beta | \beta \rangle \mp \langle \beta | -\beta \rangle]$$

$$= |\beta|^2 (1 \mp e^{-2|\beta|^2})$$

Then $g_{\pm}^{(2)}(0) = \frac{|\beta|^4 (1 \pm e^{-2|\beta|^2})}{|\beta|^4 (1 \mp e^{-2|\beta|^2})^2}$,

* Now, actually the given cat state is not normalized since $|\beta\rangle$ and $|- \beta\rangle$ states are not orthogonal.

$$\langle \Psi_{\pm} | \Psi_{\pm} \rangle = 1 \pm e^{-2|\beta|^2} \Rightarrow |\Psi_{\text{norm}}\rangle_{\pm} = \frac{1}{\sqrt{2(1 \pm e^{-2|\beta|^2})}} (|\beta\rangle \pm |- \beta\rangle)$$

→ Thus if we've done the $g^{(2)}$ calculation with the normalized state; we'd have an extra $[\langle \Psi_{\pm} | \Psi_{\pm} \rangle]^{-1}$ in the nominator and $[\langle \Psi_{\pm} | \Psi_{\pm} \rangle]^{-2}$ in the denom.; resulting in:

$$g_{\text{norm } \pm}^{(2)}(0) = \left(\frac{1 \pm e^{-2|\beta|^2}}{1 \mp e^{-2|\beta|^2}} \right)^2$$

For even cat state $|\Psi\rangle_+$, $g_+^2(0) > 1$ and thus it exhibits super-Poissonian statistics. Photons will appear bunched.

For odd cat states $|\Psi\rangle_-$, $g_-^2(0) < 1$ and thus $|\Psi\rangle_-$ exhibits sub-Poissonian statistics. Photons will appear antibunched

5) Maxwell-Boltzmann law states that, the probability of a state is related to its energy by

$$P(n) \propto e^{-E_n/k_B T}$$

For photons, n -photon state has $E_n = (n + \frac{1}{2})\hbar\omega$

$$P(n) \propto \exp[-(n + \frac{1}{2})\hbar\omega/k_B T] \propto \left(e^{-\frac{\hbar\omega}{k_B T}}\right)^n$$

$$\text{Let } e^{-\frac{\hbar\omega}{k_B T}} = \alpha \quad \text{and} \quad P(n) = C \alpha^n$$

Normalizing the prob. distribution

$$\sum_{n=0}^{\infty} C \alpha^n = \frac{C}{1-\alpha} = \frac{C}{1-\alpha} = 1 \Rightarrow C = 1-\alpha$$

Let us now calculate the mean photon number: $\langle n \rangle = N = \bar{n}$

$$\begin{aligned} N = \sum C n \alpha^n &= C \sum_{n=0}^{\infty} n \alpha^n = C \alpha \frac{d}{d\alpha} \left(\sum_{n=0}^{\infty} \alpha^n \right) \\ &= C \alpha \frac{d}{d\alpha} \left(\frac{1}{1-\alpha} \right) = C \alpha \frac{1}{(1-\alpha)^2} = \frac{\alpha}{1-\alpha} = N \end{aligned}$$

$$\alpha = 1-C \Rightarrow \frac{1-C}{C} = N \Rightarrow C(N+1) = 1 \Rightarrow C = \frac{1}{N+1}$$

Then using $C = \frac{1}{N+1}$, $\alpha = 1-C = \frac{N}{N+1}$ we find

$$P(n) = \frac{1}{N+1} \left(\frac{N}{N+1} \right)^n = \boxed{\frac{N^n}{(N+1)^{n+1}}}$$

5.b)

* Since we have the number statistics of thermal state, let us express $\langle a^\dagger a^\dagger a a \rangle$ and $\langle a^\dagger a \rangle$ in terms of number op.

$$\langle a^\dagger a^\dagger a a \rangle = \langle a^\dagger (a a^\dagger - 1) a \rangle = \langle a^\dagger a a^\dagger a \rangle - \langle a^\dagger a \rangle = \langle n^2 \rangle - \langle n \rangle$$

$$\langle a^\dagger a \rangle = \langle n \rangle = N$$

$$\langle n^2 \rangle = \sum_n n^2 P(n) = \frac{1}{N+1} \sum_{n=0}^{\infty} n^2 \left(\frac{N}{N+1} \right)^n$$

Using the derivative trick again $\Rightarrow n^2 x^n = x \frac{d}{dx} \left(x \frac{d}{dx} (x^n) \right)$

$$\langle n^2 \rangle = \frac{x}{N+1} \frac{d}{dx} \left(x \frac{d}{dx} \left(\sum_{n=0}^{\infty} x^n \right) \right) = \frac{x}{N+1} \frac{d}{dx} \left(x \frac{1}{(1-x)^2} \right)$$

$$= \frac{x}{N+1} \times \frac{(1-x)^2 + 2x(1-x)}{(1-x)^4} = \frac{x(1+x)}{(N+1)(1-x)^3} = \frac{N}{(N+1)^2} \times \frac{(2N+1)/(N+1)}{(1/(N+1))^3}$$

$$\langle n^2 \rangle = 2N^2 + N$$

$$g^{(2)}(0) = \frac{\langle n^2 \rangle - \langle n \rangle}{\langle n \rangle^2} = \frac{2N^2}{N^2} = 2 //$$

6.a) Let $|\Psi'\rangle = N a |\Psi\rangle$

Normalizing $|\Psi'\rangle$: $\langle \Psi' | \Psi' \rangle = |N|^2 \langle \Psi | \underbrace{a^\dagger a}_{\hat{n}} |\Psi\rangle = 1 \Rightarrow |N|^2 = \frac{1}{\bar{n}}$

To calculate mean photon number of the photon-subtracted state:

$$\bar{n}' = \langle \Psi' | \hat{n} | \Psi' \rangle = |N|^2 \langle \Psi | a^\dagger \underbrace{a^\dagger a}_{\hat{n}} a |\Psi\rangle$$

$$= \left(\frac{1}{\langle a^\dagger a \rangle} \right) \times \langle a^\dagger a^\dagger a a \rangle \times \frac{\langle a^\dagger a \rangle}{\langle a^\dagger a \rangle} = \frac{\langle a^\dagger a^\dagger a a \rangle}{\langle a^\dagger a \rangle^2} \cdot \bar{n}$$

$$\bar{n}' = g^{(2)}(0) \bar{n} \rightarrow \text{If initial state } |\Psi\rangle \text{ is super-Poissonian:}$$

$$g^{(2)}(0) > 1 \Rightarrow \bar{n}' > \bar{n} : \text{mean-photon number increased}$$

$\rightarrow$ If initial state is sub-Poissonian:

$$g^{(2)}(0) < 1 \Rightarrow \bar{n}' < \bar{n} : \text{mean-photon number decreased}$$

6.b) Let $|\Psi''\rangle = N a^\dagger |\Psi\rangle$

Normalizing $\langle \Psi'' | \Psi'' \rangle = |N|^2 \langle \Psi | \underbrace{a a^\dagger}_{\hat{n}+1} |\Psi\rangle = |N|^2 (\bar{n} + 1) = 1$
 $|N|^2 = 1 / (\bar{n} + 1)$

$$\bar{n}'' = |N|^2 \langle \Psi | a \underbrace{a^\dagger a}_{\hat{n}} a^\dagger |\Psi\rangle = |N|^2 \langle \Psi | (\hat{n} + 1)^2 |\Psi\rangle = |N|^2 \langle (\hat{n} + 1)^2 \rangle$$

$$\bar{n}'' = \frac{\langle (\bar{n} + 1)^2 \rangle}{\langle \bar{n} + 1 \rangle} > \langle \bar{n} + 1 \rangle : \boxed{\bar{n}'' > \bar{n} + 1} \text{ photon number is increased by at least one}$$

$\hookrightarrow$ Note that $\langle (\bar{n} + 1)^2 \rangle > (\langle \bar{n} + 1 \rangle)^2$ follows from the fact that variance is always positive $\Rightarrow E[x^2] - E[x]^2 = \text{Var}[X] \geq 0$